# How many layers does N=96 need? The last two rungs

L* = the fewest LAYERS that clear a 0.90 solve rate in EVERY depth bin at fixed hint density. Two one-change runs, both also scored on one shared 1500-instance pool.

## L*(96, ff=1024) = 8 STANDS -- but the two metrics disagree about why.

L=7 clears the 0.90 gate in 5 of 6 depth bins, missing only T=8, by 0.003.

On the pool it is +0.0007 against L=8 (92-91 discordant, p = 1): a TIE, at 88% of the parameters (5.79M vs 6.58M).

So the ladder has a rung the metric of record cannot see.

## Most of the deep-thin deficit was TRAINABILITY, not capacity.

Five epochs of LR warmup move L=12/ff=256 by +0.0800 on the pool (203-83, p = 9.1e-13).

THE CONTROL HELD: the cell with no grammar problem moved -0.0040 (p = 0.77), so warmup did not just lift the whole column.

And it is PARTIAL: the warm cell still clears the gate in 3 of 6 bins, so L*(96, ff=256) > 12 stands.

**WHAT L* IS MEASURED ON, AND THE NOTATION THE LATER PAGES USE**

Graph colouring solvable by Direct Deduction (DD) alone: colour the one node whose known neighbours leave it a single colour, repeat. N=96 nodes, degree 5, q=4 colours. An instance's DEPTH T is how many DD rounds it takes. rho = the fraction of nodes given as hints; DENSITY is the realized hint fraction, held at 0.5938 in every bin. ff = the width of the MLP inside each transformer block (ff=1024 WIDE, ff=256 THIN). A CELL is one trained model, one seed. COLD = flat learning rate, WARM = a 5-epoch ramp to the same rate. THE KNEE = L=8/ff=1024, the shallowest wide cell clearing every bin. 'Finding NN' = this project's log.

**CAVEAT, up front: these runs keep the ADJACENCY-ORDER LEAK (shuffle_adj=False), so every rate here is a DECODING score, not a deduction score. I node's neighbour row is listed sorted by the neighbours' TRUE colours, which names the node's own colour outright for 61-67% of nodes (finding 16 permuting those rows dropped the N=96 model it was measured on from 0.94 to 0.04. One seed per cell. The leak-free ladder is a separate series, r at N=48.**

Jobs 50053882 and 50008605. Prepared 2026-10-29; every number on every page is recomputed from the run outputs by this script, not typed in.

# What the model actually sees

Every token is one categorical id into a 106-symbol vocabulary (N=96, q=4, degree=5). No text, no digits-as-characters.

### VOCABULARY

| | |
|---|---|
| 0-5 | `BEGIN_HINTS BEGIN_GRAPH BEGIN_REASONING` |
| | `BEGIN_ROUND SOLVED UNRESOLVED` |
| | *markers (UNRESOLVED = DD got stuck; unused here)* |
| 6-9 | `COLOR_0 COLOR_1 COLOR_2 COLOR_3` |
| | *the q colours (token name COLOR_n, prose 'colour')* |
| 10-105 | `N0 N1 ... N95` |
| | *one id per node, no structure in the id* |

### GRAMMAR

| | |
|---|---|
| `BEGIN_HINTS` | `(NODE COLOR)*` |
| `BEGIN_GRAPH` | `(NODE (NODE){degree})*` |
| `BEGIN_REASONING` | `(BEGIN_ROUND (NODE COLOR)+)*` |

`SOLVED`

*regex: * = zero or more, + = one or more, {degree} = exactly 5*

The graph section is a neighbour table, not an image: node, then its 5 neighbours.

**THE ORDER WITHIN A ROW IS THE LEAK.**

### THIS INSTANCE, 778 TOKENS

| | |
|---|---|
| **HINTS** | `BEGIN_HINTS N0 COLOR_0 N1 COLOR_3 N2 COLOR_3 N7 COLOR_0 N8 COLOR_1 N9 COLOR_0 ... (129 tokens)` |
| **GRAPH** | `BEGIN_GRAPH N0 N81 N22 N43 N70 N10 N1 N9 N77 N52 N4 N68 ... (577 tokens)` |
| **REASONING** | `BEGIN_REASONING BEGIN_ROUND N4 COLOR_2 N13 COLOR_2 N16 COLOR_3 N21 COLOR_2 N30 COLOR_2 N33 ... (72 tokens)` |

TRAINING LOSS is next-token cross-entropy on EVERY token, the problem statement included -- there is no section mask.
AT EVAL the model is PROMPTED with everything through BEGIN_REASONING and generates the rest: 6 rounds, 32 deductions,
64 node+colour tokens, plus the BEGIN_ROUND markers and the SOLVED/UNRESOLVED terminator, which are trained targets like any other token.

DD is single-candidate elimination -- sudoku's 'naked single' rule transplanted to graph colouring, which is where the code's sudoku names come from; the task itself is the graph, not a 9x9 grid. Instance shown: seed 50000000008. These rows are the LEAKED (colour-sorted) order.

# One real N=96 instance, as the model reads it

64 hints of 96 nodes, then 6 rounds of simultaneous deduction. Hints: N0=COLOR_0 N1=COLOR_3 N2=COLOR_3 N7=COLOR_0 N8=COLOR_1 N9=COLOR_0 ...

### THE TRACE, AS EMITTED

```
BEGIN_ROUND          N4 COLOR_2 N13 COLOR_2 N16 COLOR_3 N21 COLOR_2 ...
  15 deduced
BEGIN_ROUND          N5 COLOR_3 N17 COLOR_0 N19 COLOR_3 N42 COLOR_3 ...
  6 deduced
BEGIN_ROUND          N6 COLOR_1 N22 COLOR_1 N44 COLOR_2 N52 COLOR_2 ...
  5 deduced
BEGIN_ROUND          N39 COLOR_3 N43 COLOR_3 N71 COLOR_2
  3 deduced
... 2 further rounds

SOLVED
```

### WHY ONE DEDUCTION FIRES

node N4, neighbours N60 N3 N20 N1 N24

their colours known before this round:

```
[0, 1, 3]  (3 = q-1 of the 4)

(4 known neighbours, 3 DISTINCT colours -- some repeat)

still unknown: N3
```

only colour left: 2

```
so the trace emits  N4 COLOR_2
```

*The model is never shown this reasoning -- only the node/colour pair.*

A round deduces every node whose neighbours already cover q-1 colours, all at once. Reasoning depth T is the number of rounds: a deduction in round 3 depends on ones made in rounds 1 and 2, which is the serial computation that network depth has to supply.

**THIS DECK IS ABOUT THE ROUNDS. T, the number of rounds, is the depth axis on pages 5 and 6; L* is the fewest layers that still clear 90% of instances at a given T. The adjacency-order leak lets a model skip part of that work -- see the caveat on page 1.**

T=6 rounds, the 90th percentile of 40 draws from this config (range 1-8) -- one instance, shown to make the task legible; no result in this deck is computed from it. The worked deduction on the right uses the state BEFORE this round, which is what DD itself saw; it is recomputed from the graph, not read out of the model.

# The metric of record: one pool, every comparison paired

| contrast | rates | difference | discordant (won-lost) | exact McNemar p | reading |
|---|---|---|---|---|---|
| L=7 vs L=6 -- adjacent wide rungs | 0.9120 vs 0.8833 | **+0.0287** | 126-83 | 0.00357 | resolvable |
| L=7 vs L=8 -- the knee it undercuts | 0.9120 vs 0.9113 | **+0.0007** | 92-91 | 1 | tie |
| L=12/ff256 WARM vs its COLD self -- TREATMENT | 0.8900 vs 0.8100 | **+0.0800** | 203-83 | 9.14e-13 | resolvable |
| L=8/ff256 WARM vs its COLD self -- CONTROL | 0.8367 vs 0.8407 | **-0.0040** | 137-143 | 0.765 | tie |
| L=12/ff256 WARM vs L=8/ff256 cold -- reversal | 0.8900 vs 0.8407 | **+0.0493** | 176-102 | 1.07e-05 | resolvable |
| L=12/ff256 WARM vs the wide knee | 0.8900 vs 0.9113 | **-0.0213** | 86-118 | 0.0297 | resolvable |

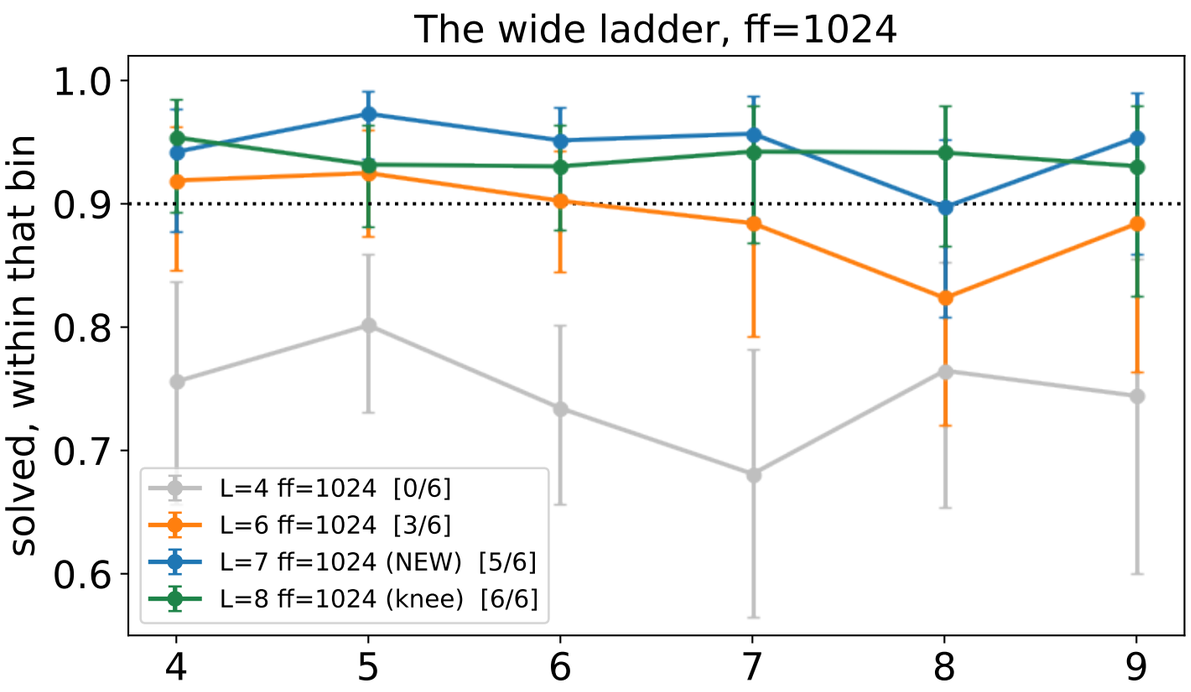
A TIE HERE IS NOT AN UNDERPOWERED TEST. L=7 vs L=8 disagree on 183 of 1500 instances and split them 92-91: the pool has the resolution and reports no difference. That is the one weakness in this ladder no further N=96 run will fix.

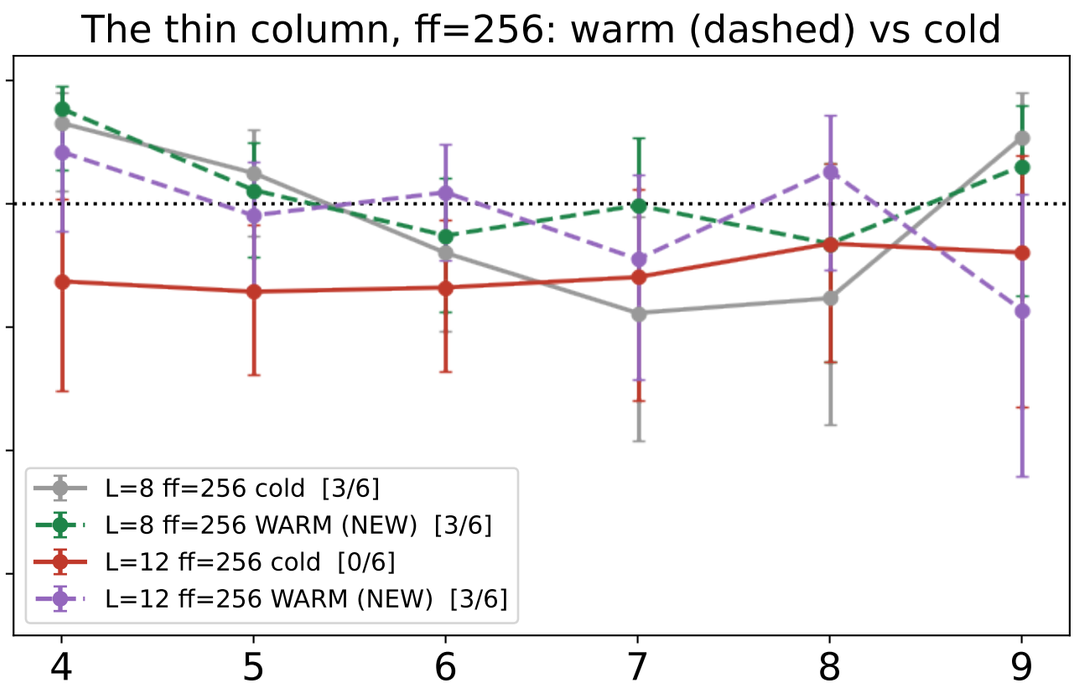
THE CONTROL IS THE LOAD-BEARING ROW. Had warmup lifted the whole thin column, finding 25's width gap would have been measured against an undertrained competitor and every N=96 width number would need re-running. The cell with no grammar problem moved by -0.0040 (p = 0.77) while the cell with one moved by +0.0800.

In the last column 'resolvable' means p < 0.05 on the paired test and 'tie' means it is not; at n=1500 a tie is a measured absence of difference, not a missing measurement. sudoku_agg_eval draws instance i at seed_start = 5e10, far above every training seed, so a larger n is a strict superset of a smaller one and all cells see byte-identical prompts: every row above is paired instance by instance. 'discordant' counts only the instances the two cells disagree on, which is what the exact test uses. The reference checkpoint's 1500-instance correctness vector is IDENTICAL across the two runs' independent eval invocations -- a free replication check.

# Where L* is read off: six depth bins, density held fixed

dotted line = the 0.90 gate;  [k/6] in each legend = how many of the six bins that cell clears



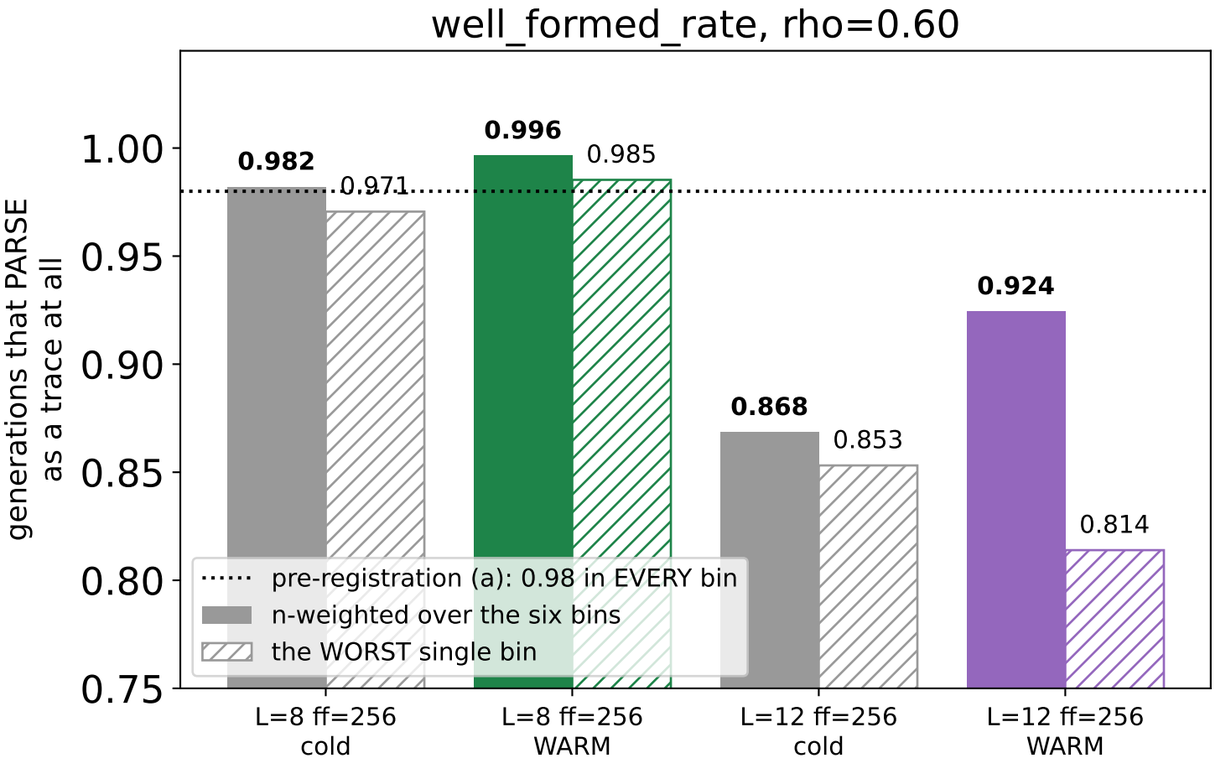




THIS IS A SEPARATE EVAL FROM THE POOL ON PAGES 4 AND 7. Here 1500 instances are drawn at rho=0.60 and binned by DD depth T, the six bins with n>=60 hold 555 of them (n = 86 (T=4), 146 (T=5), 143 (T=6), 69 (T=7), 68 (T=8), 43 (T=9)). Realized density is 0.5938 in EVERY bin of EVERY cell, so the bins differ in reasoning depth and in nothing else. The pool is a different draw, spanning the whole training range rho=0.30-0.95, and runs harder -- bin rates and pool rates are not interchangeable, which is why the two metrics can disagree about L=7.

**L=7's single miss is T=8 at 0.897 in a 68-instance bin, where ONE instance is 0.015. Three thousandths is not a measurement; it is where a threshold happened to fall -- so the N=96 width factor is quoted as '>1.5x, denominator resting on a 0.003 bin margin'.**

Cold L=12/ff=256 is FLAT and faintly RISING in T (0.829-0.868, Cochran-Armitage z = +0.67): the signature of a model whose failures are not about reasoning depth at all, which is what the grammar diagnosis on the next page predicted. Warm, it FALLS with T (z = -1.54) like an under-depth reasoner. Warmup changed the shape of the failure, not only its level.

# What warmup repaired: the model could not WRITE the trace




**THE POOL AGREES, ON ITS OWN THREE GENERATIVE STATISTICS**

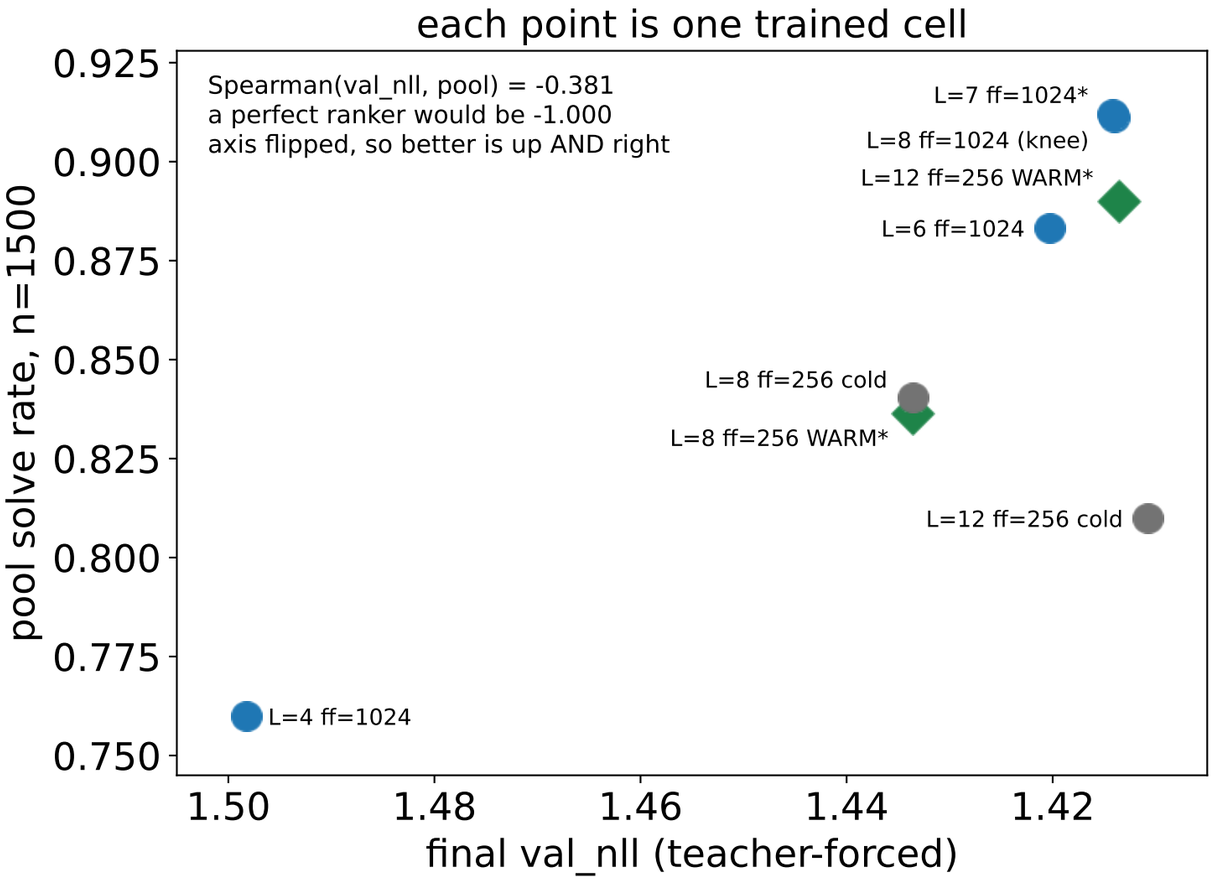
| | **treated: L=12/ff=256** | | **CONTROL: L=8/ff=256** | |
|---|---|---|---|---|
| | **cold** | **warm** | **cold** | **warm** |
| solve_rate | 0.8100 | **0.8900** | 0.8407 | **0.8367** |
| truncated_rate | 0.0167 | **0.0100** | 0.0007 | **0.0000** |
| outcome_accuracy | 0.8687 | **0.9300** | 0.9860 | **0.9953** |

Three independent generative statistics move the same way in the treated cell (green = the direction warmup was supposed to move it). In the CONTROL they barely move, so warmup did not lift the whole thin column. Finding 25's mechanism was right: most of the deep-thin deficit was the model failing to WRITE traces, not failing to deduce.

**A PARTIAL REPAIR, AND THE LIKE-FOR-LIKE PAIRS SAY SO: weighted, L=12/ff=256 goes 0.868 -> 0.924; its worst bin goes 0.853 -> 0.814, i.e. NO BETTER THAN BEFORE. An average that rises while the floor does not is a partial repair, not a fixed cell.**

well_formed_rate is the fraction of generations that parse as a trace at all -- a grammar measure, scored before any question of whether the colours are right. truncated_rate is the fraction that hit the length cap, outcome_accuracy the fraction whose SOLVED/UNRESOLVED verdict is correct. Warmup is a 5-epoch linear ramp to the same 1e-3, nothing else changed, chosen once and not scanned: 'a longer ramp would clear the gate' is untested and nothing here rules it in or out.

# The cheap metric still does not rank these architectures




**THE TWO READINGS, SEPARATELY**

**FIXED inside the thin column.**

Cold, L=12 had the LOWER nll (1.4108 vs 1.4337)

and the LOWER pool rate (0.8100 vs 0.8407) -- inverted.

Warm, it keeps the lower nll (1.4137 vs 1.4335)

and now also the higher pool rate (0.8900 vs 0.8367).

**STILL BROKEN across columns, as badly as ever.**

Warm L=12/ff=256 has a LOWER nll than the new L=7/ff=1024

(1.4137 vs 1.4142) while scoring

2.2 POINTS LOWER on the pool (0.8900 vs 0.9120).

So val_nll is not a model-selection signal at N=96, and the open

'no best-val checkpoint selection' item cannot be resolved by it.

WHY THIS MATTERS FOR THE LADDER: val_nll is what you would use to pick a checkpoint or prune a scan without paying for generation. At N=96 it would have picked the wrong cell. Every L* in this series is therefore read off generated traces, which costs about an hour of eval per cell.

Axis flipped so that BETTER is to the right on both axes. Circles are cold cells, DIAMONDS the warm-start ones, grey the cold thin column; a * marks a cell that is new in these two runs. L=7 and L=8 differ by one pixel here (1.4142/0.9120 vs 1.4141/0.9113), so their labels share a point. One seed per cell, so this is a ranking check across eight single runs, not a correlation with an error bar; it confirms finding 4 at a third task size rather than measuring a new quantity.

# Pre-registrations, as resolved

| finding | part | the claim, written down before the run was submitted | verdict | evidence |
|---|---|---|---|---|
| 31 | (a) | L=7 clears 4-5 of 6 bins, so L*(96, ff=1024) = 8 stands | **CONFIRMED** | 5 of 6; the miss is T=8 at 0.897 |
| 31 | (b) | the bins it misses are the deepest ones | **WEAK** | second-deepest, but one miss cannot trend: CA z=-1.07, p=0.28 |
| 31 | (c) | on the pool L=7 lands resolvably BELOW L=8 | **FALSIFIED** | +0.0007, 92-91, p = 1 -- a tie at 0.88x the params |
| 31 | (d) | well_formed_rate >= 0.95 in every bin | **CONFIRMED** | 0.957-0.993, weighted 0.977: no wide-column grammar failure |
| 32 | (a) | warm L=12/ff=256: well_formed >= 0.98 in EVERY bin | **FALSIFIED** | weighted 0.924; worst bin 0.814, BELOW the cold cell's 0.853 |
| 32 | (b) | given (a), it clears the gate, so L*(96, ff=256) <= 12 | **FALSIFIED** | 3 of 6 bins; L* > 12 stands, width factor still super-multiplicative |
| 32 | (c) | the control L=8/ff=256 ties its cold self | **CONFIRMED** | -0.0040, 137-143, p = 0.77 -- warmup did not lift the column |
| 32 | (d) | warmup stops val_nll ranking the thin cells in reverse | **SPLIT** | fixed INSIDE the thin column, still inverted ACROSS columns |

UNPRE-REGISTERED, and the reason this pair is worth a deck: warmup REVERSES the depth ordering inside the thin column. Cold, L=12/ff=256 scored 3.1 points BELOW L=8/ff=256 and finding 25 read that as 'depth cannot buy back MLP width at N=96'. Warm, L=12 is +0.0493 ABOVE it (176-102, p = 1.1e-05). That conclusion's direction survives -- the thin column still does not reach the wide knee -- but its sharpest sentence was about training, not architecture.

WHERE THIS LEAVES THE GRID: the MLP-width factor on L* is 1.00x / 1.50x / >1.50x at N = 24 / 48 / 96 -- super-multiplicative in N, with one confound now removed rather than suspected. WHAT IT DOES NOT ESTABLISH: leaked order throughout, so these are decoding scores; one seed per cell; the gate is sensitive at the third decimal and the pool disagrees with it about L=7 vs L=8; warmup is one untuned 5-epoch ramp; the two deepest bins carry about +-0.07 at 95%.